**Enrollment No.:** _____

Darshan UNIVERSITY

योगः कर्मसु कौशलम्

**Darshan Institute of Computer Application**

**B.Sc. (IT) | Sem-3 | Winter-2022**

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|---|---|---|---|---|---|
| **Course Code** | : | 2106CS301 | **Date** | : | 17-10-2022 |
| **Course Name** | : | Computer Network & Network Security | **Duration** | : | 150 Minutes |
| | | | **Total Marks** | : | 70 |

**Instructions:**

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

| | | | |
|---|---|---|---|
| **Q.1** | **(A)** | Draw the OSI and TCP/IP Model. | **4** |
| | **(B)** | Explain Network Layer in detail. | **3** |
| | | **OR** | |
| | | List the functions of Data-link layer. | |
| | **(C)** | Draw and Write OSI Reference Model in detail. | **7** |
| | | **OR** | |
| | | Write the Functions of the Data-Link Layer. | |
| **Q.2** | **(A)** | Show the working of a multiplexer. | **4** |
| | **(B)** | Define Multiplexing. | **3** |
| | | **OR** | |
| | | Define Switching. | |
| | **(C)** | Write a sort note on Multiplexing. | **7** |
| | | **OR** | |
| | | Write a sort note on Switching. | |
| **Q.3** | **(A)** | Calculate the broadcast address for the network 172.31.0.0/16. | **4** |
| | **(B)** | Classify IP Addresses. | **3** |
| | | **OR** | |
| | | Describe Uniform Resource Locator. | |
| | **(C)** | Demonstrate subnetting in detail. | **7** |
| | | **OR** | |
| | | Calculate the First and Last IP address of the given IP 20.10.30.35 / 27. | |

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|---|---|---|---|
| **Q.4** | **(A)** | Describe Routing Metric. | **4** |
| | **(B)** | Describe casting and list its types. | **3** |
| | | **OR** | |
| | | Clarify types of Broadcast. | |
| | **(C)** | Discuss Routing types in detail. | **7** |
| | | **OR** | |
| | | Explain Shortest Path Routing. | |
| **Q.5** | **(A)** | Importance of Network Security. | **4** |
| | **(B)** | List the useful Security Policies. | **3** |
| | | **OR** | |
| | | List the Security Methods. | |
| | **(C)** | List Firewall types and Explain anyone in detail. | **7** |
| | | **OR** | |
| | | Explain Proxy Server. | |

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